

Lecture 1: Random Vectors

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Notation

- we will use bold letters for vectors (e.g. $\mathbf{x} \in \mathbb{R}^d$) and bold capital letters for matrices (e.g. $\mathbf{X} \in \mathbb{R}^{d \times d}$)
- we will also denote random variables using capital non-bold letters, e.g. X
- random vectors will typically be denoted with bold letters, e.g. $\mathbf{x}$, unless we need to emphasize the property of being a random variable

Probability Space – definition

we begin by defining the mathematical construct that provides a formal model of a **random process**. With a **probability space**, one can model, for example, the throwing of a die.

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Definition 1 (probability space)

a probability space W is a unique triple $W = \{\Omega, \mathcal{F}, \mathbb{P}\}$, where Ω is its **sample space**, $\mathcal{F}$ its **σ -algebra of events**, and $\mathbb{P}$ its **probability measure**

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- **σ -algebra of events:** a set of outcomes in the sample space, the events A we would like to consider, i.e. subsets of Ω : $\mathcal{F} = \{A | A \subseteq \Omega, A \in \mathcal{F}\}$
- **probability measure:** a function that assigns to each event in the event space a probability, i.e. a number between 0 and 1

Example of a Probability Space – (die)

consider the following example: **throwing a standard die**

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- the **sample space** Ω is typically the set $\{1, 2, 3, 4, 5, 6\}$, where each element in the set is a label which represents the outcome of the die landing on that label

Example of a Probability Space – (die)



- The **event space** $\mathcal{F}$ could be the power set $\mathcal{P}(\Omega)$ the set of all subsets of the sample space, which would then contain simple events such as $\{5\}$ (“the die lands on 5”), as well as complex events such as $\{2, 4, 6\}$ (“the die lands on an even number”)

Example of a Probability Space – (die)



- if $\mathcal{F} = \mathcal{P}(\Omega)$, the **probability function** $\mathbb{P}$ would then map each event to the number of outcomes in that event divided by 6. For example, $\mathbb{P}(\{5\}) = 1/6$, and $\mathbb{P}(\{2, 4, 6\}) = 3/6 = 1/2$

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- We can talk about probabilities like $\mathbb{P}(X = 1)$ **without mentioning coins or outcomes anymore..**
- **Think of X as a translator:** it converts messy outcomes into numbers we can analyze.

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 - ▶ Ω : all possible outcomes;
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- A random variable X maps each outcome $\omega \in \Omega$ to a value $X(\omega)$ in a target space E .
- **Compatibility condition (measurability):**
for any “event about X ” $B \subseteq E$,

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}.$$

This ensures we can assign a probability to $X \in B$.

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- Example: for $E = \mathbb{R}$, we usually take $\mathcal{E}$ to be all intervals and combinations of intervals – the so-called **Borel sets**.

Example – random variable on a die

- $\Omega = \{1, 2, 3, 4, 5, 6\}$ (outcomes of a die)
- Define

$$X(\omega) = \begin{cases} 1, & \text{if } \omega \text{ is odd,} \\ 0, & \text{if } \omega \text{ is even.} \end{cases}$$

- Then the event $\{X = 1\}$ corresponds to $\{1, 3, 5\} \subset \Omega$.

$$\mathbb{P}(X = 1) = \frac{3}{6} = \frac{1}{2}.$$

- The key idea: **events about X correspond to sets in Ω via $X^{-1}(B)$.**

Recall Random Variables – definition

Definition 2 (random variable)

given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a **measurable space** $(E, \mathcal{E})$, a random variable is a measurable function from Ω to E , i.e. this function must satisfy the constraint $X^{-1}(B) = \{\omega : X(\omega) \in B\} \in \mathcal{F}$ for any $B \in \mathcal{E}$

Then X connects $(\Omega, \mathcal{F})$ and $(E, \mathcal{E})$:

$$(\Omega, \mathcal{F}) \xrightarrow{X} (E, \mathcal{E})$$

In words: X **transfers the probability structure from outcomes to values.**

Recall Random Variables – probability/law definition

Definition 3 (probability or law of a random variable)

the probability that a random variable X takes on a value in a measurable set $B \in \mathcal{E}$ is

$$P_X(B) := \mathbb{P}(\{\omega \in \Omega | X(\omega) \in B\}) = \mathbb{P}(X \in B)$$

Example of a Random Variable – (die)

once again consider the following example: **throwing a standard die**



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- $\Omega = \{1, 2, 3, 4, 5, 6\}$, $E = \{-1, 1\}$, $\mathcal{E} = \{\emptyset, \{-1\}, \{1\}, \{-1, 1\}\}$
- $X(\omega) = 1$ if ω is even, -1 otherwise

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- $X(\omega) = 1$ if ω is even, -1 otherwise

► take the event $B = \{1\} \in \mathcal{E} = \{\{1\}, \{-1\}, \emptyset, \{-1, 1\}\}$:

$$\begin{aligned}\mathbb{P}(X \in B) &= \mathbb{P}(\{\omega \in \Omega | X(\omega) \in B\}) \\ &= \mathbb{P}(\{\omega \in \{1, 2, 3, 4, 5, 6\} | X(\omega) \in \{1\}\}) \\ &= \mathbb{P}(\{2, 4, 6\}) \\ &= 1/2\end{aligned}$$

Definition 4 (pmf)

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- Example (infinite): $\mathbb{P}(X = i) = 2^{-i}, i \in \mathbb{N}$

Definition 5 (cdf)

define the (cumulative) **distribution function** of X by $F_X(x) = \mathbb{P}(X \leq x)$

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- F_X is always non-decreasing, right-continuous, and satisfies $F_X(-\infty) = 0$, $F_X(+\infty) = 1$.
- if X is **discrete**, the CDF is **stepwise** and you can formally write $F_X(x) = \sum_{x_i \leq x} P_X(x_i)$

Absolute Continuity and Density

Definition 6 (Absolutely Continuous Random Variable)

A random variable X is said to be **absolutely continuous** if there exists a nonnegative, Lebesgue-integrable function f_X such that

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx \quad \text{for all } a < b.$$

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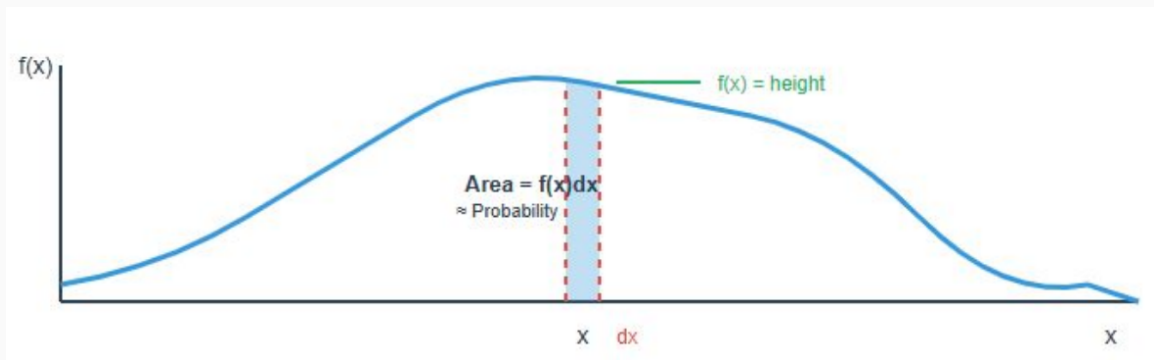
Connection with the CDF

If X is absolutely continuous, then

$$F_X(x) = \int_{-\infty}^x f_X(t) dt \quad \text{and hence} \quad f_X(x) = \frac{dF_X(x)}{dx} \quad (\text{almost everywhere}).$$

PDF and intuition

intuitively, one can think of $f_X(x) dx$ as being the **probability** of X falling within the infinitesimal interval $[x, x + dx]$



Expectation of a Random Variable – definition

Definition 7 (expectation)

if a random variable X has density $f_X(x)$, then its expectation is given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

Expectation of a Random Variable – discrete case

for a discrete random variable with a finite list $x_1, \dots, x_k$ of possible outcomes each of which (respectively) has probability $p_X(x_1), \dots, p_X(x_k)$ of occurring, the expectation can also be written as

$$\mathbb{E}[X] = \sum_{i=1}^{\infty} x_i p_X(x_i).$$

Expectation of a Random Variable – linearity (part 1)

Proposition 1 (linearity of expectation)

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

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proof: let $f_{X,Y}(x, y)$ be the joint pdf of X and Y . By definition, the expectation of $X + Y$ is given by:

$$\begin{aligned}\mathbb{E}[X + Y] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f_{X,Y}(x, y) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X,Y}(x, y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{X,Y}(x, y) dx dy\end{aligned}$$

the marginal pdf of X (resp. Y) is given by $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$ (why?)

(resp. $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$)

Expectation of a Random Variable – linearity (part 2)

Proposition 2 (linearity of expectation)

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

the first term of the sum

$$\mathbb{E}[X + Y] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X,Y}(x, y) dx dy + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{X,Y}(x, y) dx dy$$

becomes

$$\int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \right) dx = \int_{-\infty}^{\infty} x f_X(x) dx = \mathbb{E}[X]$$

and the second integral becomes

$$\int_{-\infty}^{\infty} y \left(\int_{-\infty}^{\infty} f_{X,Y}(x, y) dx \right) dy = \int_{-\infty}^{\infty} y f_Y(y) dy = \mathbb{E}[Y]$$

we have: $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$

□

Expectation of a Random Variable – tail integral

Proposition 3

the expected value of a non-negative random variable can be written as the integral of its tail distribution:

$$\mathbb{E}(X) = \int_0^{+\infty} \mathbb{P}(X \geq t) \, dt$$

proof: this is because if X is a nonnegative real number, then for each realization $x = X(\omega)$,

$$x = \int_0^{+\infty} 1_{\{x \geq t\}} \, dt$$

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we integrate both sides of the identity with respect to the distribution dF_X of X and use Fubini theorem to change the order of the summation/integral and of the expectation:

$$\begin{aligned} \mathbb{E}(X) &= \int_0^{+\infty} x \, f_X(x) \, dx = \int_0^{+\infty} \int_0^{+\infty} 1_{\{x \geq t\}} \, dt \, f_X(x) \, dx = \int_0^{+\infty} \int_0^{+\infty} 1_{\{x \geq t\}} f_X(x) \, dx \, dt \\ &= \int_0^{+\infty} \int_t^{+\infty} f_X(x) \, dx \, dt = \int_0^{+\infty} \mathbb{P}(X \geq t) \, dt \end{aligned}$$

Example: Expectation of an Exponential Random Variable

as an example, we can use this identity to compute the expected value of an **exponential** random variable without having to integrate by parts

Example: Expectation of an Exponential Random Variable

- let X be a standard exponential random variable with density

$$f_X(x) = \begin{cases} e^{-x}, & x \geq 0, \\ 0, & x < 0 \end{cases}$$

- instead of computing

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx \quad (\text{it requires integration by parts})$$

we compute

$$\mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X \geq t) dt$$

- we have for any $t \geq 0$ that $\mathbb{P}(X \geq t) = \int_0^t e^{-x} dx = e^{-t}$

$$\text{then } \mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X \geq t) dt = \int_0^{\infty} e^{-t} dt = 1$$